IM-1 ORBIT DETERMINATION: A COMPARISON OF MISSION DELIVERIES VS POST-FLIGHT TRAJECTORY RECONSTRUCTION

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Intuitive Machines' (IM) inaugural mission, IM-1, was launched on 2/15/2024 at 06:05:00 UTC on a SpaceX Falcon 9 from Kennedy Space Center. The IM-1 spacecraft, a Nova-C class lander named Odysseus, landed near the Malapert A crater on 2/22/2024 at 23:23:00 UTC. Odysseus used radiometric tracking data from IM's Lunar Telemetry Network, supplemented by short-duration passes from the Deep Space Network, to successfully navigate into lunar orbit and then descend to the surface of the Moon. This paper summarizes the operation experiences, navigation results, and lessons learned of the IM and KinetX Ground Navigation teams and compares the "as flown" trajectory to the post-mission reconstructed trajectory.

INTRODUCTION

Intuitive Machines (IM) recently met its goal of softly landing a robotic spacecraft near the south pole of the Moon in February 2024 with its inaugural mission, IM-1. The IM-1 mission was catalyzed by NASA's Commercial Lunar Payload Services (CLPS) program which aims to further the exploration of the lunar surface in preparation for human missions to the Moon. The success of IM-1 was the first time in over 50 years the United States landed a spacecraft on the Moon, as well as the first time a private company was able to accomplish this feat.

One of many innovative features of the IM-1 mission was its spacecraft navigation approach. The IM-1 navigation team, which consisted of KinetX's Space Navigation and Flight Dynamics (SNAFD) group in partnership with IM's Ground Navigation team, used radiometric tracking data (e.g., two-way Doppler, two-way range, and three-way Doppler) to successfully navigate the IM-1 spacecraft, Odysseus, in trans-lunar space and in lunar orbit. This tracking data was primarily supplied by IM's Lunar Telemetry Network (LTN) and the Kongsberg Satellite Services (KSAT) Network making the IM-1 mission the first mission to use a commercial tracking network to navigate in deep space. Furthermore, Odysseus's navigation was supplemented by several short-duration – <30 min – passes from the Deep Space Network (DSN) for navigation validation during key points in the IM-1 trajectory. Innovation is inherently risky, and IM-1 was no exception. This

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paper summarizes the operational experiences of the IM-1 navigation team, the navigation lessons learned from the IM-1 mission, and compares Odysseus's "as flown" navigation solutions to the post-mission reconstructed trajectory.

MISSION OVERVIEW (IM - 3 PAGES)

The IM-1 mission was launched on 2/15/2024 at 06:05:00 UTC on a SpaceX Falcon 9 from Kennedy Space Center and landed near the Malapert-A crater on 2/22/2024 at 23:23:00 UTC.

Mission Conops

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Spacecraft Overview

The IM-1 Nova-C lander, also called Odysseus, is a hexagonal cylinder with six landing legs and is 3.938 meters (12.98 feet) tall. It has a launch mass of 2000.9 kg with a dry mass of 677.7 kg. The vehicle used a gimbaled VR900 main engine composed of methane and oxygen as liquid propellants pressurized by helium gas. The vehicle used a helium reaction control system (RCS) for attitude control.

Lunar Tracking, Telemetry, and Command Network Overview

IM ground station network called Lunar Tracking, Telemetry & Communications Network (LTN) is composed of a network of ground stations and Nova Control to receive and consolidate the data. S-band was used for IM-1.

IM provided the half rack and configured each BBU for each ground station IM partnered with directly. Each one had a different configuration.

- DSS17 (Morehead) and GHY6 (Goonhilly) used an Amergint S-band BBU connected directly to the uplink and downlink feeds. The BBU itself managed all frequencies.
- DSS49 (Parkes) is a receive-only dish. It used a 70Mhz Amergint BBU. Channel one was connected to a downconverter tuned to the nominal Thales transmit frequency and the other to the nominal Quasonix downlink frequency.
- OKN2 (Okinawa) used a 70Mhz Amergint BBU. This was connected to the station uplink chain and the station downlink chain, including site-provided up and down conversion.
- HBK1 and HBK2 (Hartebeesthoek) used a 70Mhz Amergint BBU in two types of configurations. In both configurations, the uplink signal is routed into a fiber converter and routed to the HBK2 (12 meter) dish, where it is converted back into RF, upconverted, amplified and transmitted. Reception is either via the HBK1 (29 meter) or HBK2 (12 meter). In either case, the signal is received, downconverted to the 70Mhz IF, fed into a fiber converter, routed to the equipment room with the BBU, converted back into RF, and fed into the BBU.

In addition to IM's existing sites, IM also utilized the Norwegian company Kongsberg Satellite Services (KSAT)'s network of ground stations. These devices are similar to the 70 MHz units deployed to IM LTN sites but with a different software buildout. Most ground stations are capable

	LDN Sites	Dish Size (m)	S-band	X-band	Ka-band
Existing/Near Term	Parkes	64	↓	↓	-
	Hartebeesthoek	12/13.2/26	↕	↕	↕
	Goonhilly	32	↕	↕	-
	Morehead	12/21	↕	↕	↓
	Mopra	22	↓	↓	↓
Future	Santa Paula	18.3	↕	↕	↕
	Goonhilly	18.3	↕	↕	↕
	Dongara	18.3	↕	↕	↕

	Partner Sites	Dish Size (m)	S-band	X-band	Ka-band
SSC	Punta Arenas	7.3/11.5		↕	↓
	Inuvik	7.3		↕	↓
	Bangkok	7.3		↕	↓
	Kiruna	7.3		↕	↓
KSAT	Kourou	15	↕	↓	
	Mauritius	7.3	↕	↓	
	Dubai	7.3	↕	↓	
	Singapore	9.1	↕	↓	
	India	18/32	↕	↓	

Figure 1. Map of the ground stations. Note that IM used only Parkes, Hartbeesthoek, Goonhilly, and Morehead as part of its LTN network. Parkes was the only dish IM utilized that only has downlink capabilities. Six different KSAT dishes were used.

of both uplink and downlink but a few are only capable of downlink. The ground station network provides a seamless coverage across the world as can be seen in Figure 1.

Each of these ground stations have a minimum elevation angle relative to Earth's local horizon and maximum range. Prior to the mission, using our nominal trajectory and ground station limitations presented a visibility schedule with a custom GMAT script. This visibility schedule was then processed through an internally developed tool called the Dish Scheduling Tool which optimized the tracking, telemetry and command coverage to meet mission requirements while minimizing cost, risk and complexity. This was then passed on to Operations personnel for final manual refinement by humans to ensure that all dishes are utilized and usage contracts with them properly fulfilled.

On the spacecraft itself, Odysseus employed two types of radios (two of each) – a Thales Alenia transponder and a Quasonix transmitter. The Thales transponders were effectively identical whereas the Quasonix were of two varieties – Quasonix 1 was a 25 Watt transmitter and Quasonix 2 was a 20 Watt transmitter. The Thales units had a nominal transmit frequency of 2,210,600,000.00Hz and a nominal receive frequency of 2,035,594,166.66Hz. This is with a turnaround ratio of 240/221. The uplink signal was a center carrier surrounded by an 8,000Hz command subcarrier and a 100,000Hz ranging subcarrier. The downlink signal was a center carrier surrounded by a 30,144Hz telemetry subcarrier and a 100,000 Hz ranging subcarrier or a BSPK signal modulated directly onto the carrier. The tone set employed had minor tones of 20000Hz, 16800Hz, 16160Hz, 16032Hz, 16008Hz, and 16000Hz. The Quasonix units are largely software defined; they were set for a nominal downlink frequency of 2,250,000,000Hz with a variety of data rates.

The lander employed two types of antennas. It has four low gain hemispherical turnstile antennas and one high-gain dish approximately 15 cm in diameter. Both types of antennas used RHCP. Any radio could be routed to any antenna via the onboard switch fabric, but the number of hops and the amount of cable between each radio and antenna varied. Additionally, there as a filter placed on the

HGA that precluded proper Thales operation – this was due to a fear of self-jamming.

During the mission, tracking data was gathered in real time from the ground sites via the native APIs provided by the baseband units. For Cortex Units, three elemental measurements are taken – Doppler, Integrated Phase, and Ranging. These are each delivered from the API as arrays of measurements tagged with an initial timestamp and a sampling rate frequency. Each of these measurements are recorded to the database with Remote Time generated from the time tag, plus the measurement index times the sampling rate.

For Amergint Units, two elemental measurements are taken – Integrated Phase and Ranging. Integrated Phase is delivered as a timestamped value, and recorded verbatim. Instantaneous Doppler is extracted from the integrated phase by subtracting the previous integrated phase measurement and normalizing time range to 1 second, which gives the equivalent of an instantaneous measurement in delta Hz/ second. Ranging is delivered as a timestamped value and is recorded verbatim.

Ultimately, the telemetry could be downlinked and saved in real-time via an internally built data query tool which saves all the associated data in json format. This was then processed via another tool to generate the timestamped range and range rate data passed on to the OD team.

ORBIT DETERMINATION

KX/IM (Jeroen) - 6 pages

The orbit determination (OD) solutions throughout the 1-week mission used the Double Precision Orbit Determination Program (DPODP) also known as Mirage and a predecessor to the Jet Propulsion Laboratory (JPL)'s MONTE. The program is based on the pseudo-epoch state filter algorithms developed by Bierman[1] (1977, republished by Dover in 2006). The primary tracking measurements incorporated into the OD solutions were two-way, and three-way Doppler as well as ranging. The majority of these measurements came from IM's ground tracking network discussed in subsection "Lunar Tracking, Telemetry, and Command Network Overview" with a few Deep Space Network (DSN) tracking passes incorporated where it was deemed necessary.

Measurement Models

KX/IM

The process by which the measurements were ingested in DPODP was circuitous but simply because DPODP was originally meant to be the back-up OD tool to the General Mission Analysis Tool (GMAT). Consequently, measurement tracking files were generated in the range and range-rate GMAT format. These files were then re-converted back into the original Doppler and range measurements using the nominal uplink frequency and put in a Tracking Data Message (TDM) file format which was then converted to a binary NIO file read by DPODP.

All non-DSN tracking passes were on S-band for both uplink and downlink. The nominal conversion between frequency and range-rate is 135.62 mm/s/Hz for 2-way Doppler and 6.79 RU/m for 2-way range.

Dynamical Models and Filter Setup

KX/IM

The latest JPL development ephemeris 440 (DE440) was implemented for this mission, and as per the recommendation from the Solar System Dynamics (SSD) group, we continued to use the

DE430 correlated covariance and DE430 partials. For the Earth, the Grace Gravity Model 02C (GGM02C) was implemented with degree and order 20. For the Moon, the Grail Gravity Model 1200B (GRGM1200B) was implemented with degree and order 90. The spacecraft had an initial wet mass of 2034 kg [Shen has mass of 2000.9 kg instead].. and utilized a simple Solar Radiation Pressure (SRP) cannonball model because there was no attitude profile.

Table 1. Orbit Determination Filter Configuration

Parameter	Type	A-priori 1-σ	Notes
Epoch State	Estimated	1$\times$ nominal LVSEP Covariance	State and uncertainty obtained from LVSEP
SRP Scale Factor	Estimated	0.2	-
Per-Pass Range Biases	Estimated	15 m	White noise stochastic model, batch start at range pass
Accelerations	Estimated	1.0E-10 km/s^2	Per axis, white noise stochastic model with 3 hour batches
Earth Ephemeris	Considered	12$\times$12 DE430 Earth-Venus Covariance	SET III parameters
Station Locations	Considered	10 m	non-DSN Assets
Troposphere	Considered	0.01 m / 0.01 m	Dry/Wet
Ionosphere	Considered	0.55 m / 0.15 m	Day/Night for X-band
Pole X and Y	Considered	0.002 m	-
UT1	Considered	0.025 m	-

Mission Orbit Determination Solutions

KX/IM

During the 7-day mission, 8 OD solutions were delivered. In this section we discuss what tracking data and navigation assumptions went into these solutions which are presented on a B-plane plot. For completeness the the launch time was Thursday 15-Feb, 2024 00:05 CST. The first OD was delivered on Thursday 15-Feb 16:45 CST to characterize the launch injection trajectory as well as for the design of the fixed-dV magnitude Commissioning maneuver (CM). Using the LVSEP covariance to constrain the possible lunar trajectories, two tracks of 30 minutes of Doppler data were incorporated into the OD001 solution. The first track was from Singapore (SI1) and the second track was from Byalalu, India (D32). When mapping the OD solution out to the LOI mapping time as shown in Figure 2 we see that even 1 hour of tracking data significantly collapses the uncertainty from the previous LVSEP covariance.

The second OD, OD002, was delivered on Friday 16-Feb 18:00 CST. This OD solution was a pre-CM solution and contains Byalalu, India (D18) and D32 Doppler tracking data as well as near-continuous ranging data from Goonhilly, UK (GHY6), Morehead, KY (DSS17), and Okinawa, Japan (OKN2). The execution of the CM was delayed by 24 hours allowing more tracking prior to the maneuver execution which occurred on 16-Feb 19:17:02.920 CST, shortly after the OD002 delivery.

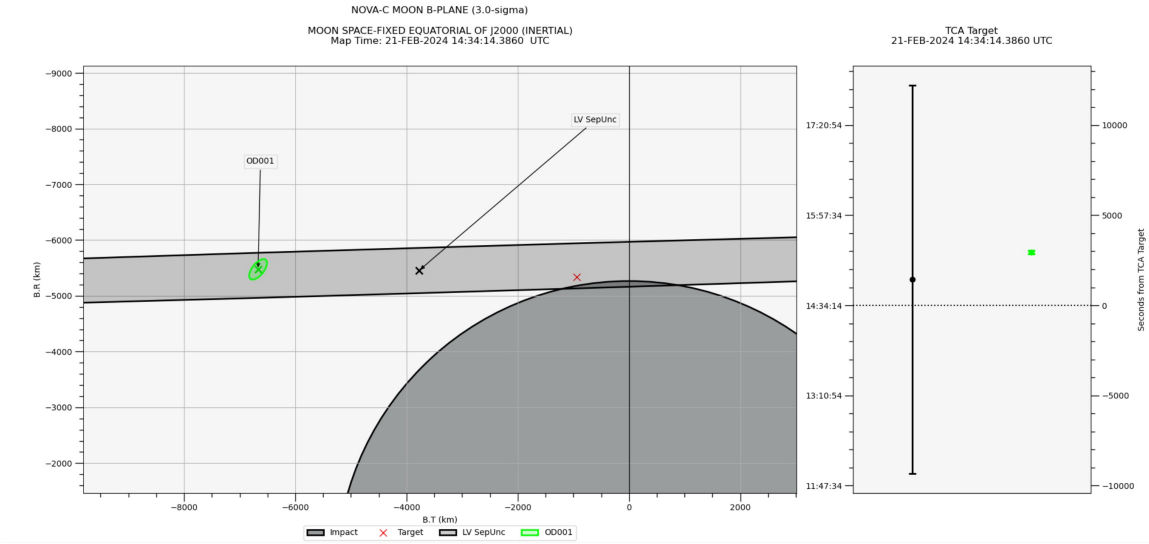

Figure 2. OD001 Mapped to LOI.

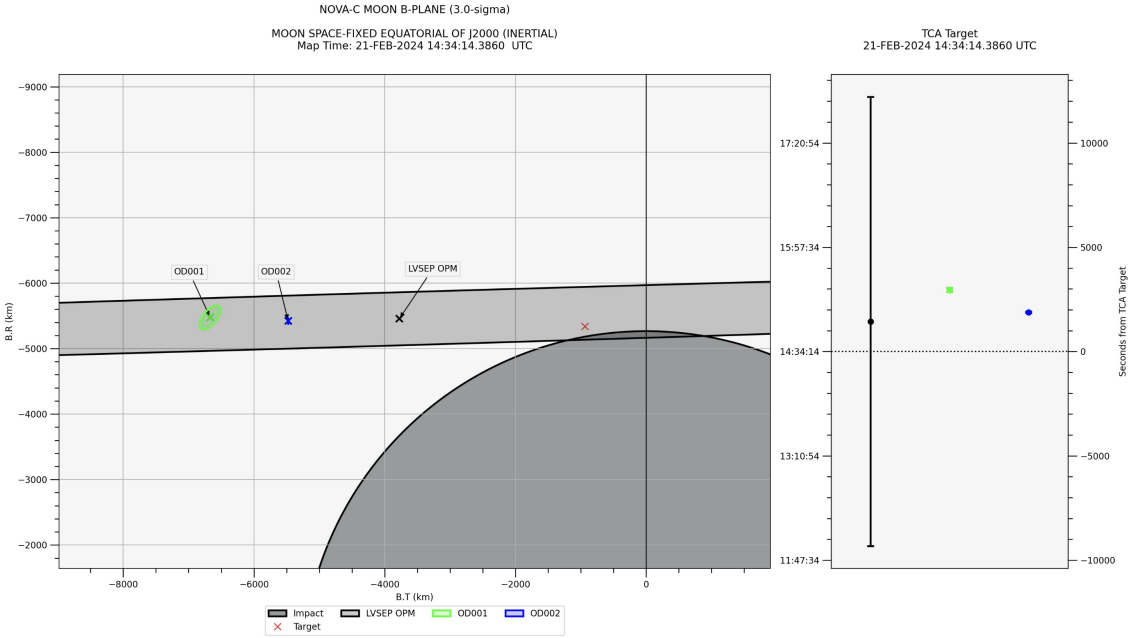

Figure 3. OD002 Mapped to LOI.

From Figure 3 it is clear that the OD solution has moved significantly between OD001 and OD002. This movement in the b-plane is either as a result of inconsistent tracking data between the various tracking stations which is causing differing results, or due to unmodeled accelerations. These accelerations could include the residual dV from the Reaction Control Thruster (RCS) firings to maintain the spacecraft attitude, which are unmodeled and unknown at the time of the IM-1 mission. Other large unmodeled accelerations such as a helium leak could also significantly alter the predicted trajectory. Smaller errors in modeling such as using a cannonball model for SRP, certainly

will have an impact as well, but these would be smaller than the previously mentioned errors.

The subsequent OD, OD003 was delivered on Saturday 17-Feb 12:30 CST and was the first solution where the LVSEP covariance constraining the solutions no longer had a large weight in the solution due to the significant maneuver execution errors introduced from the CM.

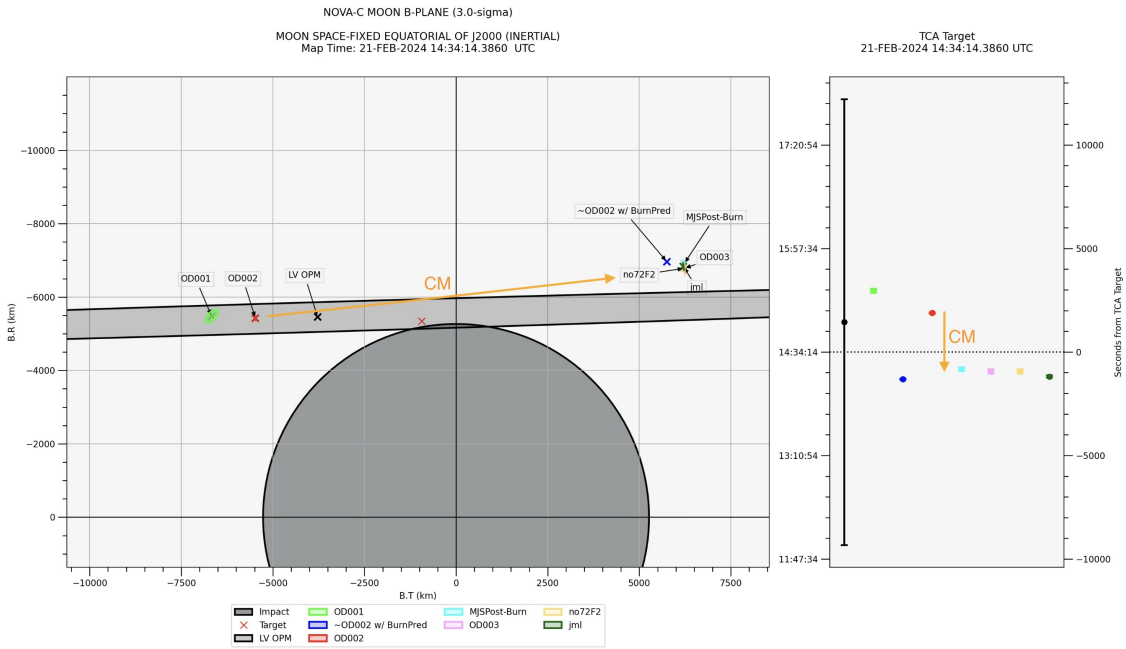

Figure 4. OD003 Mapped to LOI.

In Figure 4 the operational b-plane from OD003 and its other candidate solutions indicate the large shift in the b-plane introduced from the execution of the CM which was nominally 21 m/s. The blue $\times$ in Figure 4 indicates where we would have expected to end up in the B-plane had the nominal CM in both dV and pointing been executed exactly. However, incorporating the tracking data at hand from OD002, and adding in a Doppler pass from GHY6 after the CM (this Doppler pass also had a large bias estimated) in addition to ranging data from GHY6, DSS17, and OKN2 provided an overburn estimate of CM moving it farther along the B·T direction in the b-plane. The other b-plane solutions near OD003 are the various other candidate solutions analyzed with varying the filter input parameters.

Table 2. Commissioning Maneuver Formal Reconstruction

Epoch (UTC)	Parameter	Design	Estimated	Smoothed σ	A-priori σ
17-Feb-2024 01:17:02.920	ΔV (m/s)	21.0	21.07	0.093	1.0
	RA (deg)	306.538	304.793	0.333	1.0
	DEC (deg)	-2.348	-4.503	0.636	1.0
	Burn Start (sec)	0	-6.152	0.697	1.0

Table 2 shows the estimated parameters of the CM from OD003. The dV and pointing are estimated to be within the expected range, however the estimated start of the burn was 6 seconds earlier than expected check with BET if the same result still holds. OD003 was used for the design of TCM-1 which was designed to execute on 18-Feb 20:30 CST.

OD004 was delivered on Monday 19-Feb 18:45 CST and used a post CM epoch and included the reconstruction of TCM-1 which had an anomalous early cutoff. This OD solution was also the first to incorporate a limited amount of DSN tracking data. The post-CM Doppler included was from DSS17, as well as 20 minutes from the DSN Goldstone station. Ranging measurements in this solution were included from GHY6, DSS17, OKN2, and Hartebeesthoek, South Africa (HBK1).

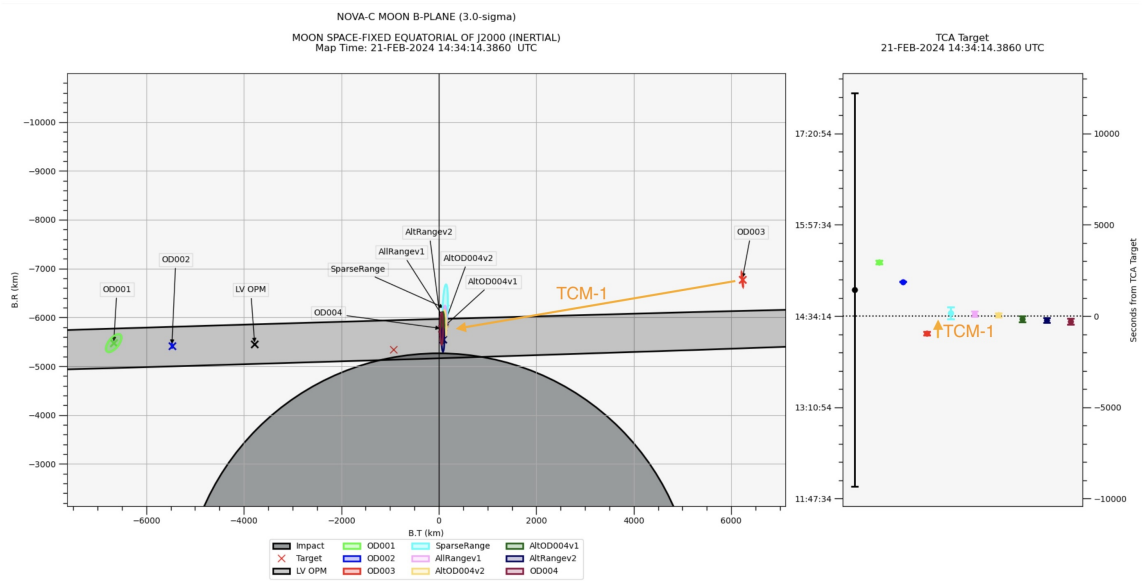

Figure 5. OD004 Mapped to LOI.

From Figure 5 we see that the TCM-1 maneuver did not put the spaceraft over the LOI target designated on the b-plane by the red target $\times$. This was a result of the early cutoff of TCM-1. Table 3 illustrates the actual achieved performance of the TCM-1 burn.

Table 3. TCM-1 Formal Reconstruction

Epoch (UTC)	Parameter	Design	Estimated	Smoothed σ	A-priori σ
19-Feb-2024 02:30:00	ΔV (m/s)		27.09	0.221	3.0
	RA (deg)		130.071	0.478	7.1
	DEC (deg)		1.452	1.167	7.1
	Burn Start (sec)	0	3.655	0.494	1.0

Why doesn't the OD005 + TCM3 bplane coincide with the target in Figure 6?

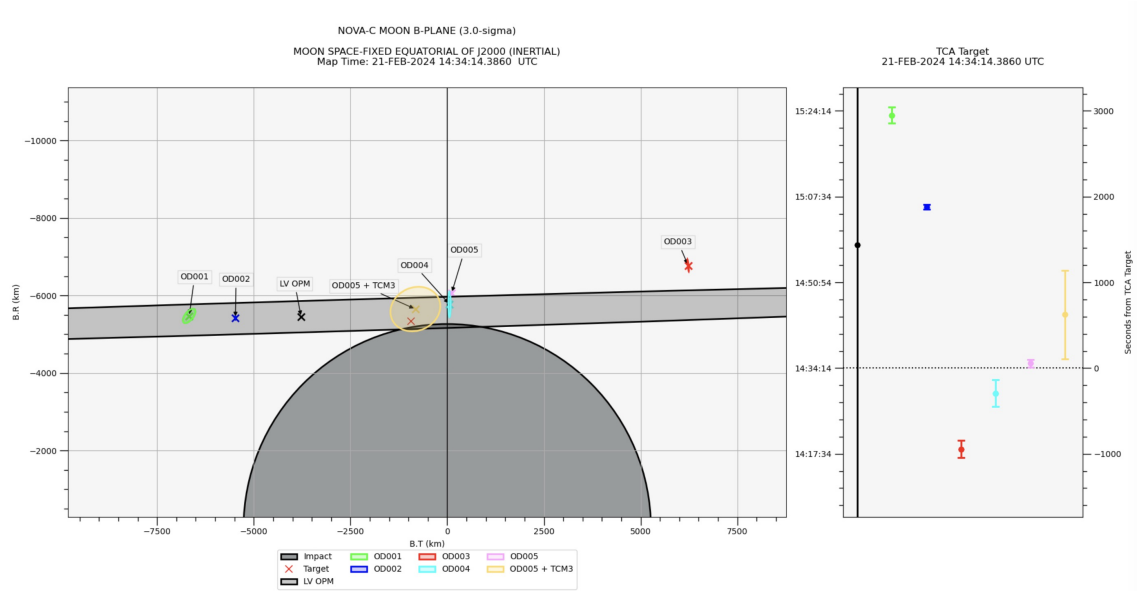

Figure 6. OD005 Mapped to LOI.

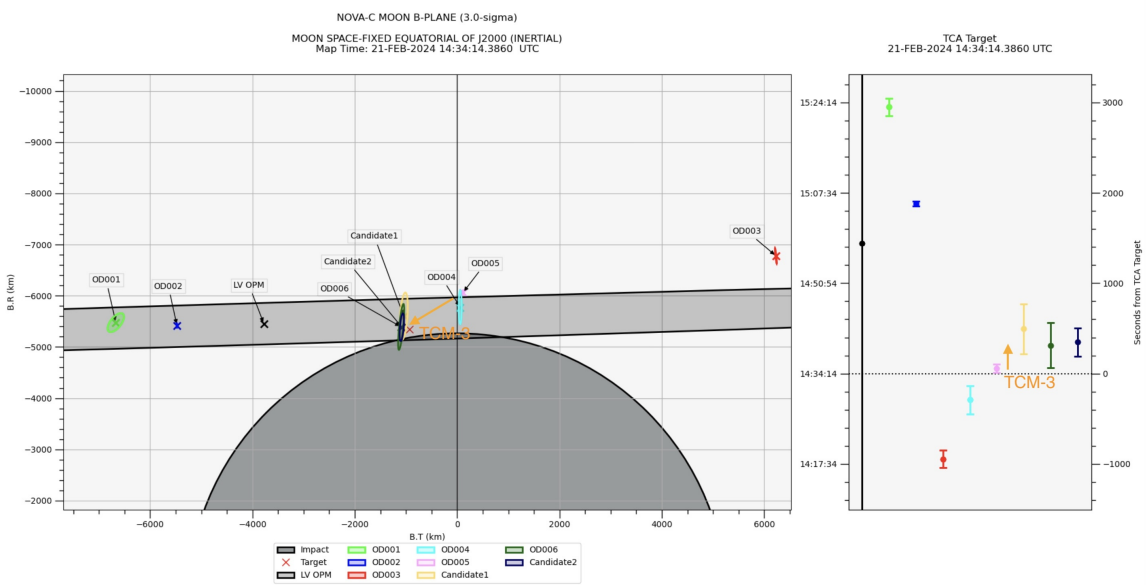

Figure 7. OD006 Mapped to LOI.

DISCUSSION OF MISSION OPERATION DIFFICULTIES

IM/KX - 4 pages

Tracking Data/Communication Issues

IM/KX (David/Jason)

Table 4. TCM-3 Formal Reconstruction

Epoch (UTC)	Parameter	Design	Estimated	Smoothed σ	A-priori σ
19-Feb-2024 02:30:00	ΔV (m/s)		19.123	0.935	3.0
	RA (deg)		130.071	0.478	7.1
	DEC (deg)		1.452	1.167	7.1
	Burn Start (sec)	0	3.655	0.494	1.0

Maneuver Execution Error Issues

IM/KX

RCS/Attitude Issues

IM/KX

POST-MISSION BEST ESTIMATED TRAJECTORY

Limited contemporaneous knowledge of spacecraft activities and sparse tracking data has made the need for a Best Estimated Trajectory (BET) great. In general, the work that has been done in the months following landing can be summarized as two efforts: to improve the spacecraft modeling, and to recover as much usable tracking data as possible.

Spacecraft Modeling Improvements

Given the limited time between KinetX involvement in the IM-1 mission and expected launch window, a simplified model of the spacecraft was implemented. Both KinetX and Intuitive Machines implemented a spherical cannonball SRP model, with an area of 4.0 m^2. This also removed the need for an accurate attitude profile, since the accelerations observed by this cannonball model would be attitude independent. It was expected that there would be mismodeling as a consequence, however, the Scale Factor estimate would allow the filter room to adjust accordingly. For the BET, KinetX implemented a ray-traced model, a high-fidelity shape model that has been implemented successfully on other missions (CITE OREX and/or LUCY). Ray-traced models have been observed to improve SRP predictions two-fold (Cite AGU Fall 2019) as compared to plate models, and are expected to improve predictions even more as compared to the cannonball model.

To effectively use the ray-traced model, historical attitude information must be used in the integration process. While no historical data of this kind was provided during operations, the ground system (?) team was able to parse through the spacecraft telemetry and provide a SPICE C-Kernel with a data span from just after launch to shortly before landing. This, with the rest of the SPICE kernels, provided a full history of spacecraft orientation data which was processed and using with in-house KinetX software.

With the physical spacecraft model improved, the modeling of spacecraft events was next. Since IM-1 had an RCS attitude control system, any errors in thruster alignment would result in a translational momentum exchange that must be modeled. During operations, RCS thruster firing errors were modeled via stochastics, since they could not be predicted in advance, there was sparse or indefinite reconstruction of firings epochs, and there was only insufficient or unreliable tracking data to estimate them. The Ground System/Propulsion Team (?) was able to provide a time-history of cumulative thruster firings, which KinetX processed, shown in Figure 8 below.

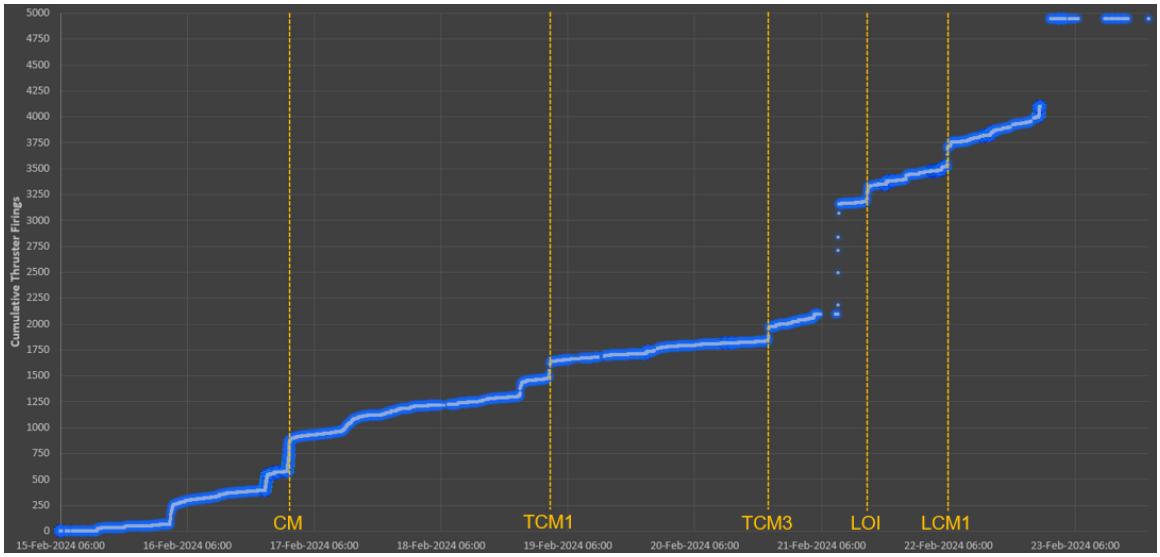

Figure 8. IM-1 Reconstructed Cumulative Thruster Firings. A time history of the total thruster firings recorded via telemetry is plotted, along with vertical lines at maneuver epochs.

While it is easy to correlate when there are deterministic maneuvers, there are periods of high thrusting which need to be understood. To better see where these periods happen, it is useful to find at the rate of thrusting by taking the change in cumulative thrusting over each period of time, or elapsed time between records. This results in a simple linear slope, $delta$thrust/$delta$time, which works well for high-rate records. Most of the time history data is recorded at 1Hz. During times where there is greater elapsed time during records, we see in Figure 9 that the rate is not affected much since the cumulative thruster firings are still counted properly following gaps in telemetry.

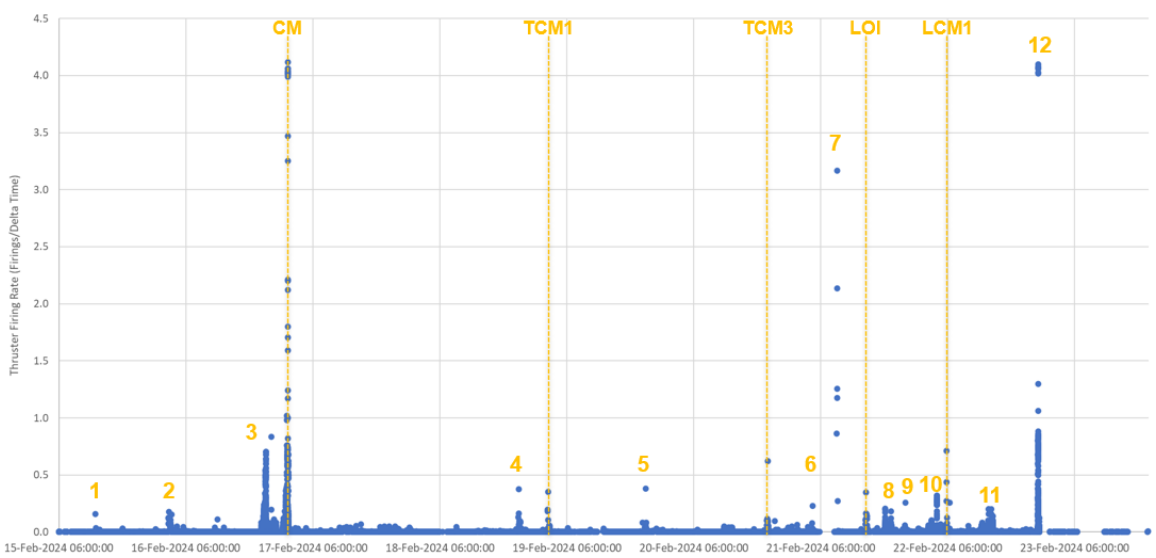

Figure 9. IM-1 Thruster Firing Rate. A rolling time history of the thruster firing rate is plotted. Events with rates exceeding 0.15 firings/second are noted here.

Since part of this exercise is to understand what happened during operations, an arbitrary metric

of 0.15 firings/second is used to note special events. As expected, there are high rates of thruster firings around maneuvers, as well as during the spacecraft resets (?) and in particular the CPU reset that occurred just before LOI. During that time, the spacecraft was tumbling at an angular rate upwards of > 1.0 degrees/second (?), and high rates of RCS thrusting was needed to recover and keep the spacecraft back to within nominal dead-bands.

To this point, all that is known are total thruster firings and inferred rates of thruster firings. The effects of misalignment and resulting residual Delta-V are still not characterized, however, given some assumptions we can create upper and lower bounds for accelerations due to misalignment observed by the spacecraft during each phase of the mission. By taking an aggregate approach, we can calculate the total Delta-V imparted to the spacecraft from RCS thrusting via the Ideal Rocket Equation, and assume some percentage of that is exchanged as linear momentum:

$$Delta - V = Isp * g0 * \log(\frac{mf}{me}) \tag{1}$$

where:

$$\begin{aligned} DV &= Delta - V(m/s) \\ Isp &= SpecificImpluse(s) \\ g0 &= accelerationduetogravityonEarth \\ mf &= EmptyMass + PropellantMass \\ me &= EmptyMass \end{aligned}$$

This form of the Rocket Equation assumes that each RCS thruster firing is short, therefore the effect of gravity on the maneuver can be ignored. For many individual maneuvers, the Delta-V scales linearly. Given an ISP of 135s for each nozzle and the constant g0 = 9.80665 m/s^2, the Delta-V can be found.

Improvements for IM2

IM (Sam)

CONCLUSION

IM (Don) - 1 page

IM-1's success was not only a step towards laying the foundation for a sustainable presence on the lunar surface, but also represents a pivotal moment in the ongoing narrative of space exploration.

REFERENCES

[1] G. J. Bierman, *Factorization Methods for Discrete Sequential Estimation*. Dover, 2006.